Walker AI - Room Measurement to Quantity Engine

Developer handoff for converting room dimensions into estimating quantities with minimal bottlenecks.

Purpose

This document defines how Walker AI should convert room measurements into base estimating quantities such as floor tile SQF, wall tile SQF, drywall SQF, framing LF, trim LF, paintable area, and fixture counts. The goal is to give developers a simple, debuggable rules layer that works with manual dimensions first and can later accept LiDAR or photo-derived measurements.

Core Formula

Dimensions -> Quantity Rules -> Base Pricing Library -> Adjustment Engine -> Permit / Risk Layer -> Estimate Range

Input Source	MVP Support	Later Upgrade
Manual dimensions	Yes	Keep
Simple room wizard	Yes	Keep
LiDAR scan	Optional	Yes
Photo recognition	No	Later

MVP rule: build the quantity engine from rectangular room math first. Do not block launch waiting for perfect scan automation.

1. Required Inputs

Field	Type	Notes
Room length	Feet + inches	Interior measurement
Room width	Feet + inches	Interior measurement
Ceiling height	Feet + inches	Use actual measured height
Door count	Integer	Subtract openings where needed
Window count	Integer	Subtract openings where needed
Wet walls / shower walls	Selection	Needed for wall tile and waterproofing
Fixture selections	Selection	Toilet, vanity, tub, shower, fan, GFI, lights
Bulkheads / soffits present	Yes/No	Affects drywall and framing

2. Standard Conversions

Rule	Formula
Floor area	length x width
Perimeter	2 x (length + width)
Gross wall area	perimeter x ceiling height
Ceiling area	length x width
Opening deduction	sum of door and window areas
Net wall area	gross wall area - opening deduction

Assume standard defaults only when the user does not know exact sizes. Defaults must be editable.

Default	Value
Standard door area	21 SQF
Standard window area	15 SQF
Standard vanity width	24 to 60 inches based on user selection
Standard shower footprint	32x48 or 36x60 unless custom

3. Quantity Rules by Trade

Trade Item	Quantity Rule	Notes
Floor tile	Use net floor area x waste factor	Waste factor 8% to 12%
Wall tile	Selected tile wall area x waste factor	Only tiled walls
Shower wall tile	Sum of shower wall rectangles x waste factor	Use shower height
Drywall walls	Net wall area not tiled	Subtract tile areas and openings
Drywall ceiling	Ceiling area	Add bulkheads separately
Paintable walls	Net wall area not tiled and not full-height millwork	Subtract cabinets if kitchen
Baseboards	Perimeter - door widths - full shower openings	Output in LF
Casing	Door count x standard casing LF + window count x standard casing LF	Output in LF
Wall framing	New wall length in LF	User or designer selects new walls
Bulkhead framing	Bulkhead length in LF	Separate line item
Waterproofing floor	Wet floor area	Bathrooms or shower zones
Waterproofing walls	Wet wall area	Shower / tub surround only

Recommended waste factors

Material	Default Waste
Tile	10%
Large format tile	12%
Laminate / vinyl / hardwood	7% to 10%
Drywall sheets	8%
Paint	5%

4. Room-Type Logic

Room Type	Primary Quantities	Conditional Triggers
Bathroom	Floor tile, wall tile, waterproofing, fan, GFI, vanity, toilet	Plumbing relocation, drain move, vent penetration, permit
Kitchen	Cabinet LF, backsplash SQF, flooring SQF, lights, GFI, appliances	Gas line, electrical upgrade, plumbing relocation
Basement finish	Framing LF, insulation SQF, drywall SQF, ceiling SQF, flooring SQF	Moisture, egress, structural, permit
Laundry / mudroom	Flooring SQF, paint SQF, GFI, venting, plumbing	Drain move, gas, vent penetration

Bathroom quantity example

Input	Example
Room size	5 x 8 feet
Ceiling height	8 feet
Door count	1
Window count	1
Shower tile walls	3 walls totalling 84 SQF

Output Quantity	Formula	Result
Floor area	5 x 8	40 SQF
Gross wall area	2 x (5+8) x 8	208 SQF
Opening deduction	1 door (21) + 1 window (15)	36 SQF
Net wall area	208 - 36	172 SQF
Shower wall tile with 10% waste	84 x 1.10	92.4 SQF
Floor tile with 10% waste	40 x 1.10	44 SQF
Baseboards	26 LF perimeter - 3 LF door	23 LF

5. Conditional Logic and Flags

Trigger	Engine Action
House age before 1960	Raise hidden-condition risk and show asbestos / old plumbing caution
Basement project	Show basement condition, bulkhead, moisture, and low ceiling logic
Plumbing relocation = yes	Add plumbing adjustment and show drain move question
Drain move + basement = yes	Use slab relocation pricing and high-risk weighting
Not sure count >= 3	Expand estimate range and recommend site review
Structural wall change = yes	Flag expert review and permit review

6. Output Schema for Developers

Output Field	Example
floor_area_sqf	40
wall_area_net_sqf	172
shower_tile_sqf	92.4
floor_tile_sqf	44
baseboard_lf	23
drywall_ceiling_sqf	40
waterproofing_sqf	84 to 92.4 based on waste rule
fixture_counts	{toilet:1, vanity:1, fan:1, gfi:2}
flags	[basement_project, permit_review, low_confidence]

7. Implementation Notes

Store formulas, defaults, and waste factors in config tables rather than hard-coding them into UI screens. That keeps the engine editable when pricing changes. Start with rectangular room math, then layer in custom-shape support later by splitting rooms into rectangles.

Recommended build order: (1) manual dimensions wizard, (2) quantity outputs, (3) pricing hookup, (4) adjustment rules, (5) confidence / risk engine, (6) LiDAR integration.